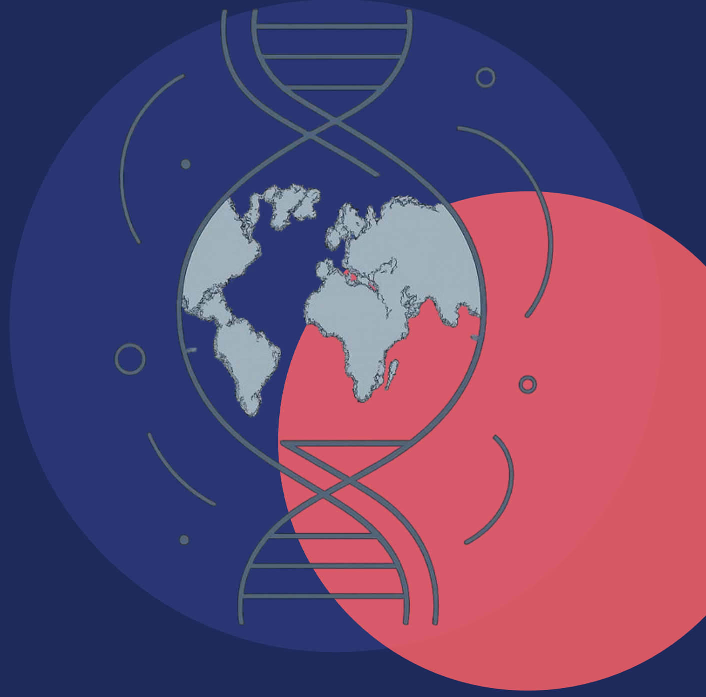
TYPE 2 DIABETES • POLYGENIC RISK SCORES • HEALTH EQUITY

A Secondary Analysis: Cross-Ancestry Transferability of European Type 2 Diabetes Risk Scores to South Asian Populations



Kim Luong, B.S.

McWilliams School of Biomedical Informatics, UTHealth Houston

BMI 6313: Scientific Writing in Healthcare | Dr. Lex Frieden

A Genetic Weather Forecast for Disease

STEP 1

Scan the genome

Studies compare the DNA of thousands of people to find spots linked to a disease.

STEP 2

Add up the signals

A polygenic risk score sums thousands of small genetic effects into one number.

STEP 3

Forecast the risk

The goal: flag high-risk people years before disease develops, so they can act early.

A Disease Growing Fastest Where We've Studied It Least

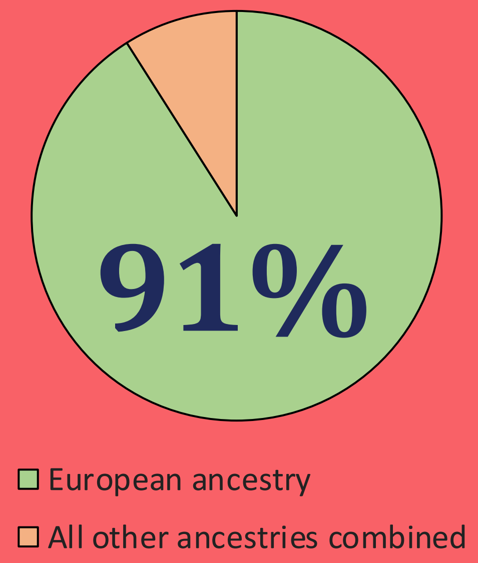
Polygenic risk scores (PRS) could flag high-risk individuals years before diabetes onset, but the discovery data behind them is deeply skewed.

783M

people projected to have type 2 diabetes globally by 2045
(Hossain et al., 2024)

Central and South Asian populations carry some of the highest rates of type 2 diabetes in the world — yet remain among the least represented in the genetic studies meant to help predict it.
(Fitipaldi & Franks, 2022; Ding et al., 2023)

Ancestry of GWAS Participants



Central & South Asian populations are among the least represented in this data — despite carrying the highest disease burden. *(Ding et al., 2023)*

CSA Underrepresented: *Most vulnerable to T2D, least studied in genomic research*

Drivers of Cross-Ancestry Failure

Allele frequency differences

How common a disease-linked variant is can differ across ancestries, so a score calibrated on European data can misjudge genetic risk elsewhere. *(Mahajan et al., 2022)*

LD structure mismatch

Nearby variants are inherited together in clusters (haplotypes), and these patterns differ by population — so European index variants can miss the true causal variants elsewhere.

70–80% of cross-ancestry PRS accuracy loss is attributed to these two factors in prior research

(Kachuri et al., 2024; Wang et al., 2020)

But for South Asians, What's Really Driving It?

1

Different Biology

Disease architecture genuinely diverges across ancestries

2

Technical Artifacts

Mismatched linkage-disequilibrium (LD) patterns and allele frequencies

3

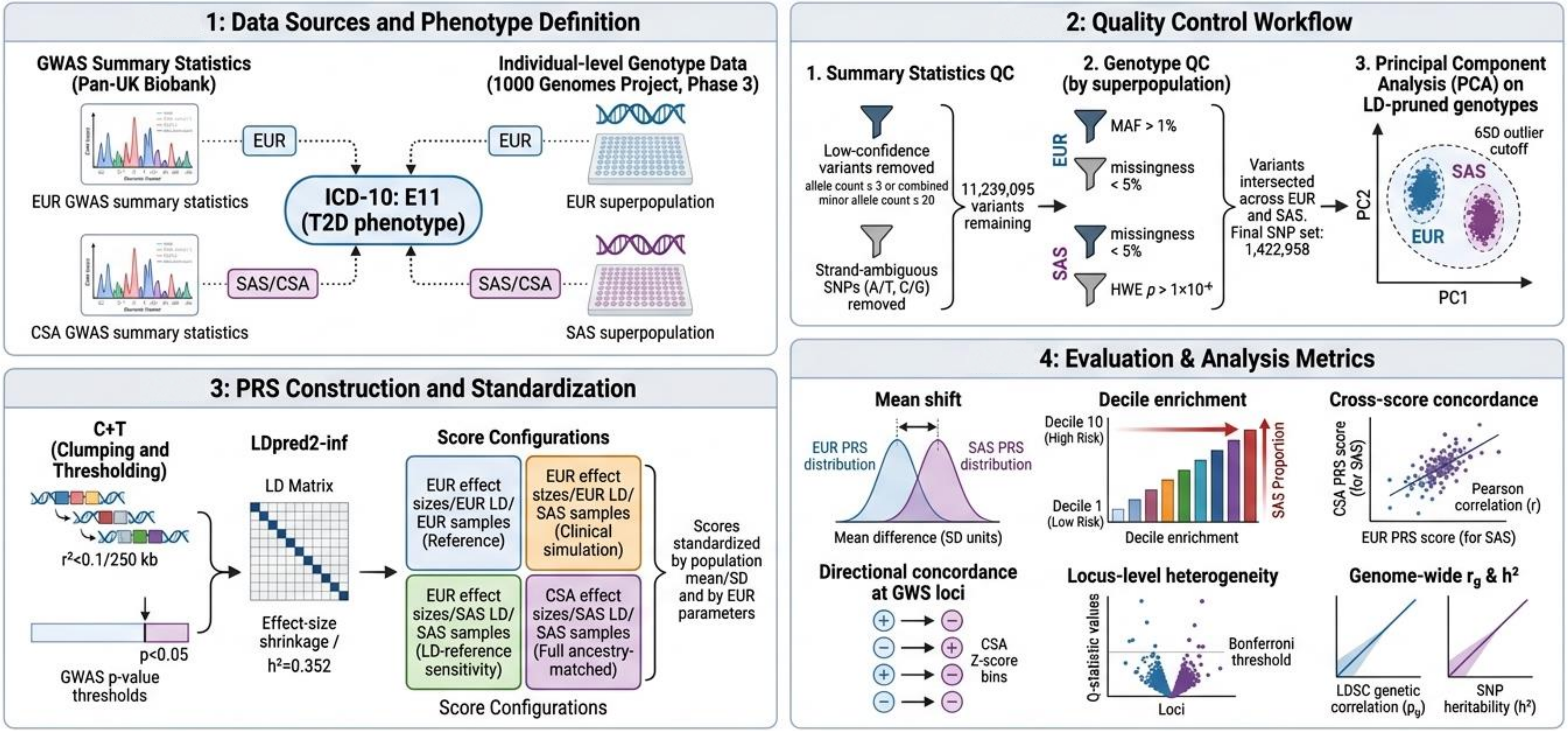
Limited Discovery Power

Simply not enough South Asian GWAS data to estimate effects precisely

Hypothesis: Limited discovery power in South Asian cohorts — not divergent disease biology — is the dominant driver of poor PRS transferability. The distinction matters: the fix differs completely depending on which is true.

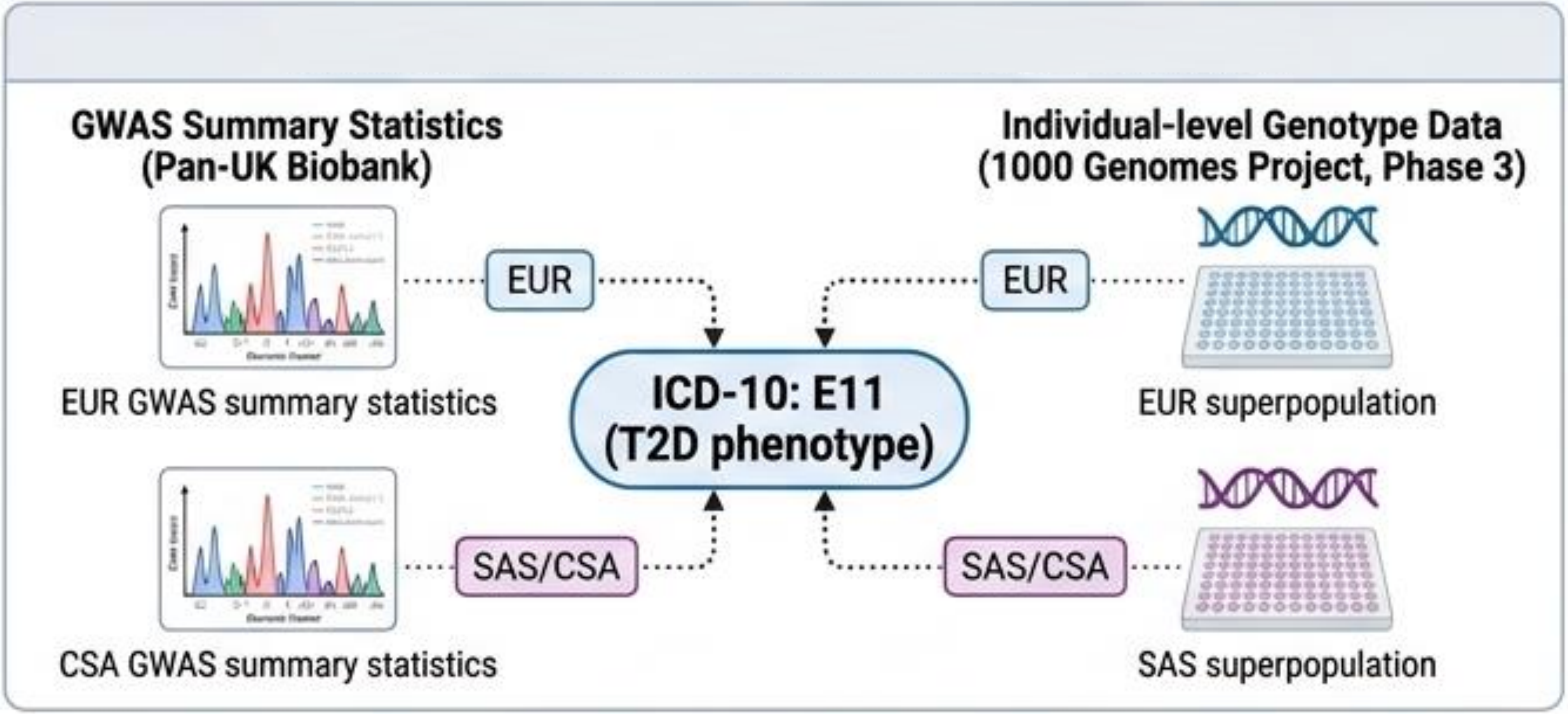
Study Design at a Glance

Public summary statistics + individual genotypes → QC & ancestry PCA → four PRS configurations → five transferability m etrics



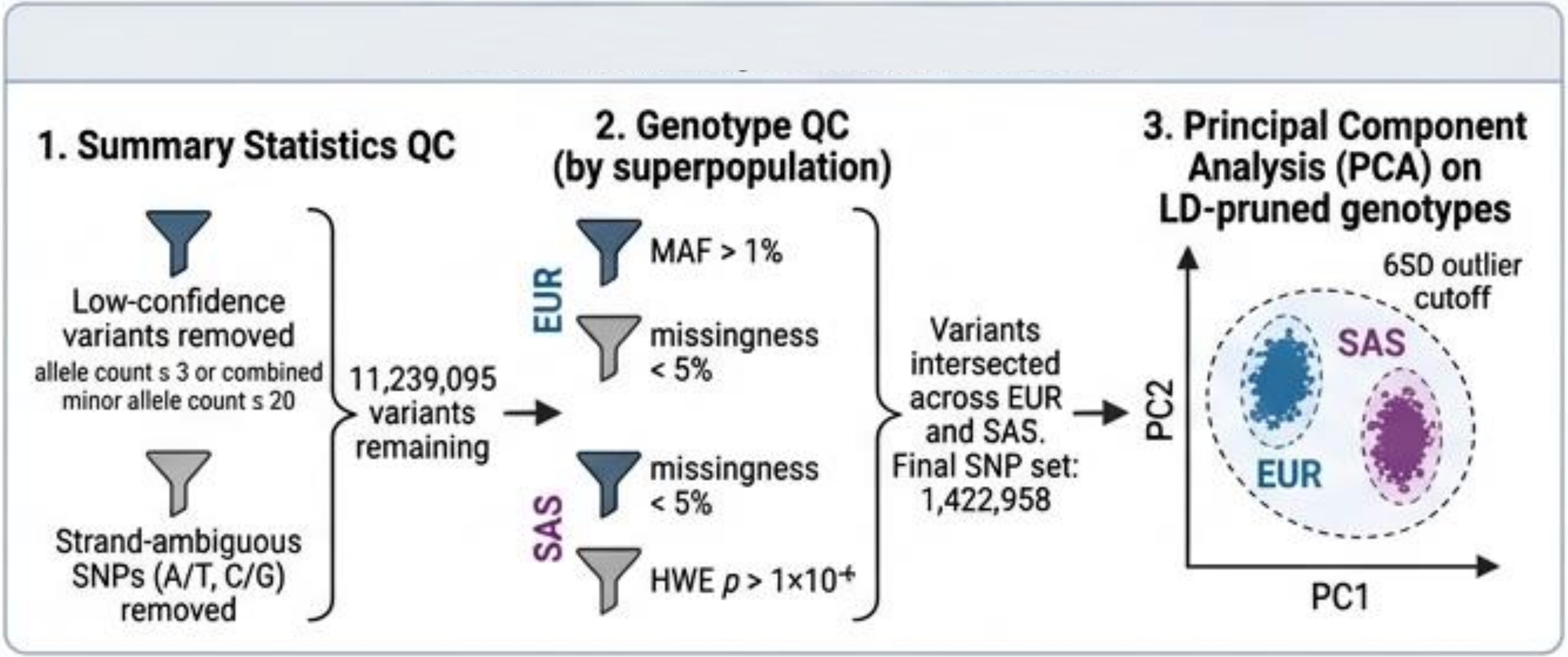
1. Data Sources and Phenotype Definition

Public summary statistics + individual genotypes → QC & ancestry PCA → four PRS configurations → five transferability metrics



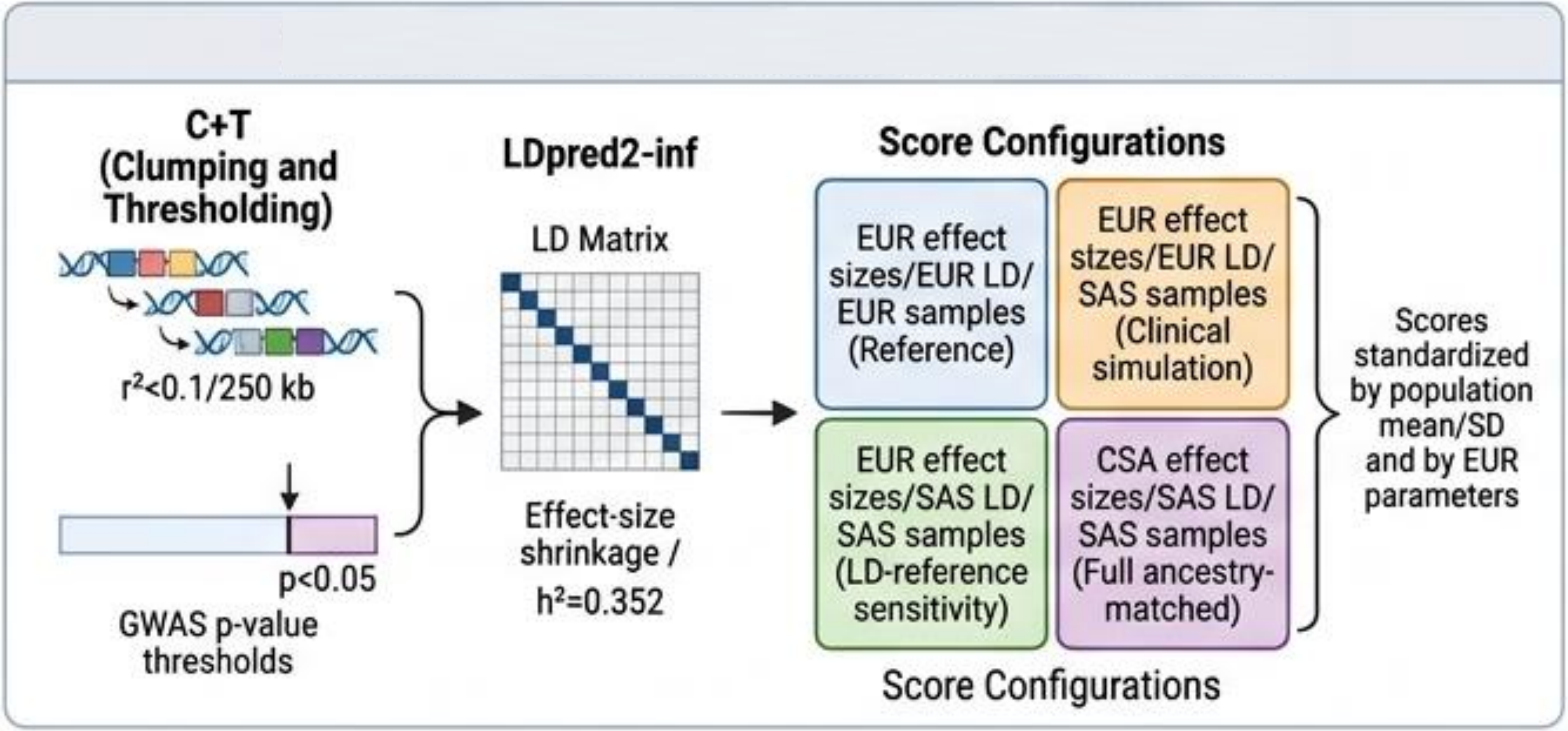
2. Quality Control Workflow

Public summary statistics + individual genotypes → QC & ancestry PCA → four PRS configurations → five transferability metrics

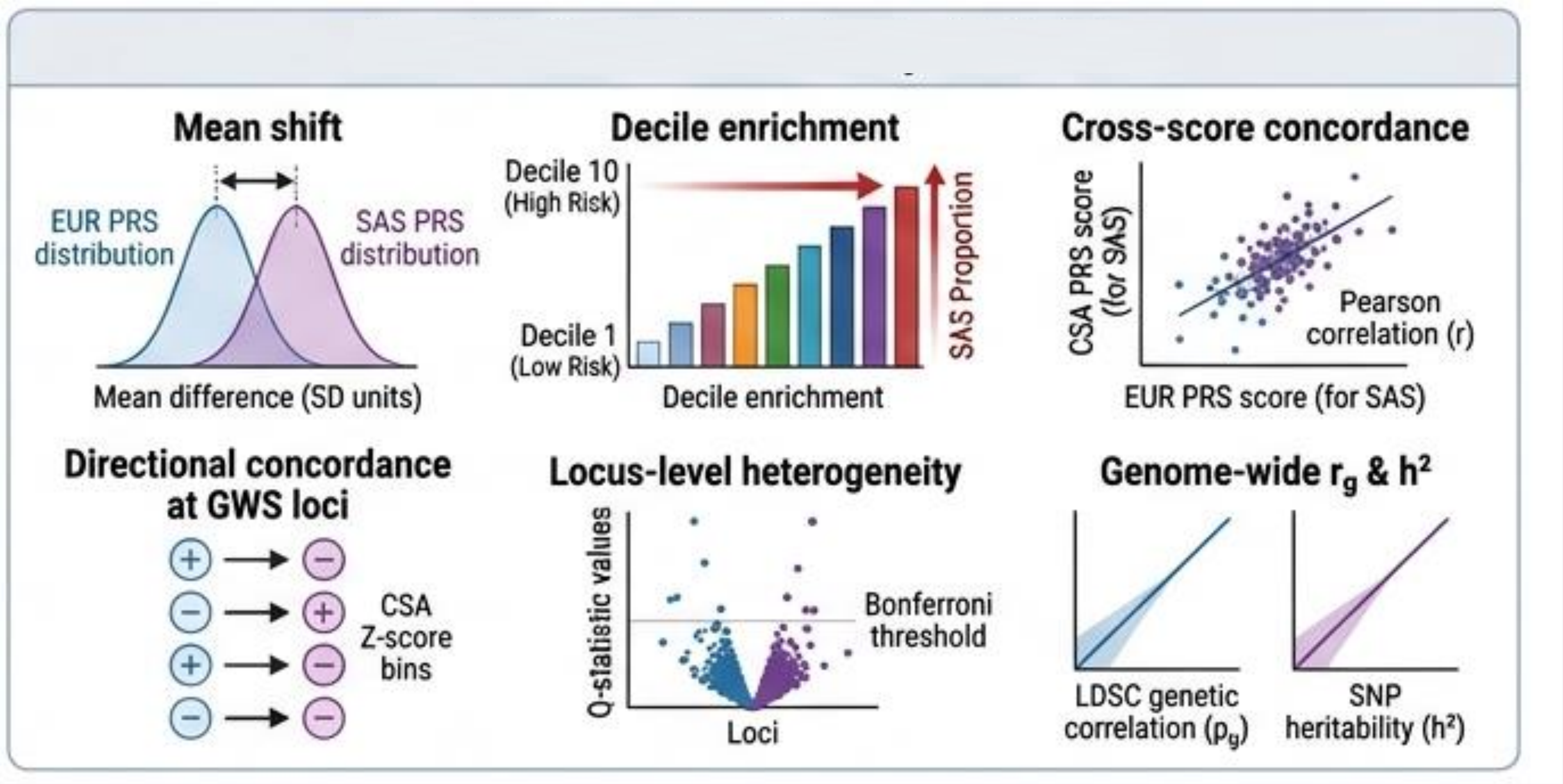


3. PRS Construction and Standardization

Public summary statistics + individual genotypes → QC & ancestry PCA → four PRS configurations → five transferability metrics



4. Evaluation and Analysis Metrics



The Limiting Factor Is Discovery Power, Not the Scoring Method

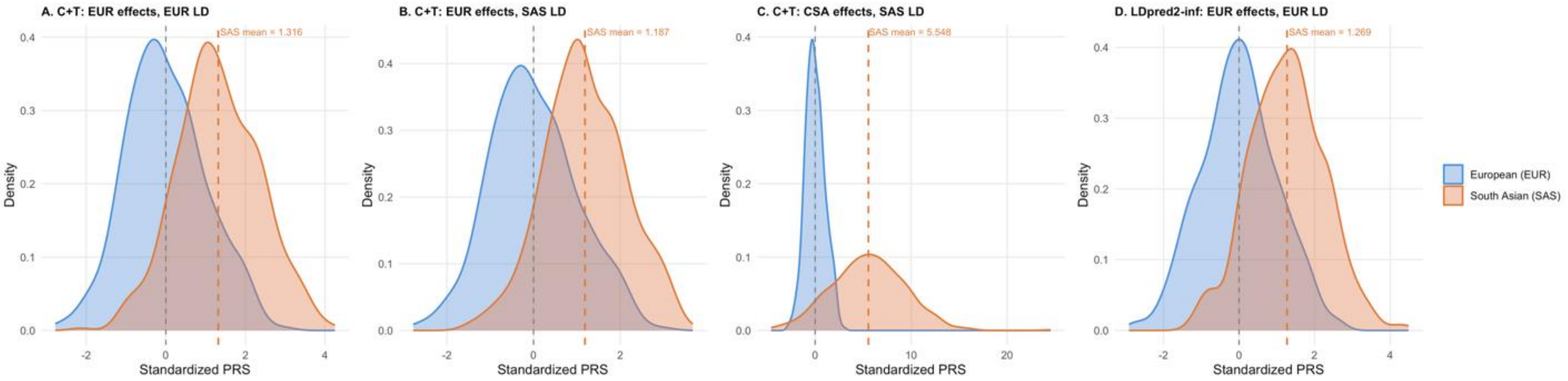


Figure 2. Density distributions of standardized PRS across score configurations.

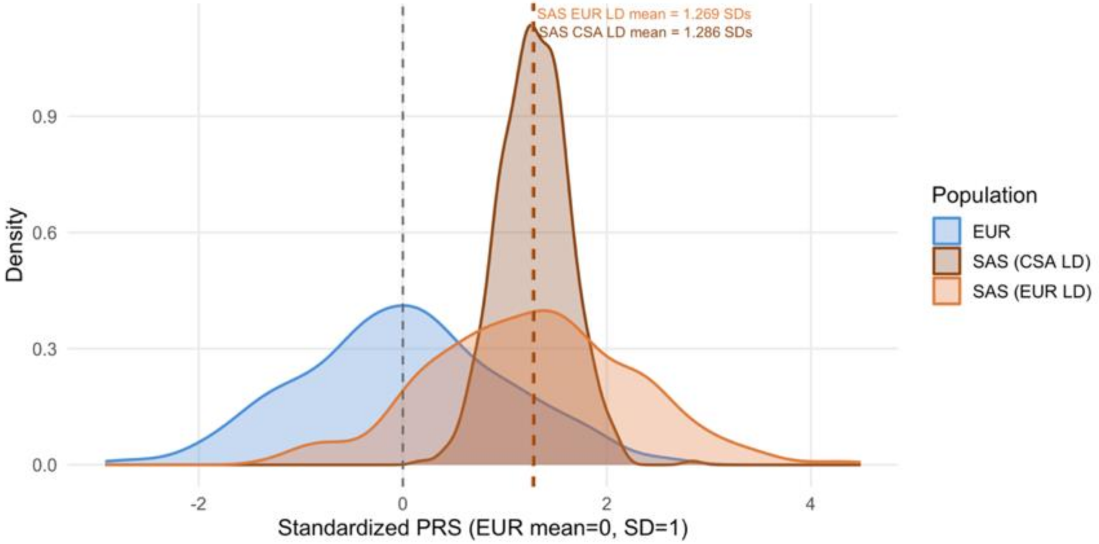


Figure 3. LDpred2-inf PRS density across ancestry groups.

+1.3 SD upward shift in South Asian scores under both methods (C+T: 1.32 SD, LDpred2-inf: 1.27 SD)

1.32 → 1.19 SD *Swap LD reference only modestly narrowed the shift .*

1.19 → 5.55 SD *Swap effect-size source made the shift larger and far noisier*

0.34 vs 0.98 SD *Shift stays ~1.29 SD, but the distribution collapses instead of widening.*

Clinical Risk Stratification Is Miscalibrated

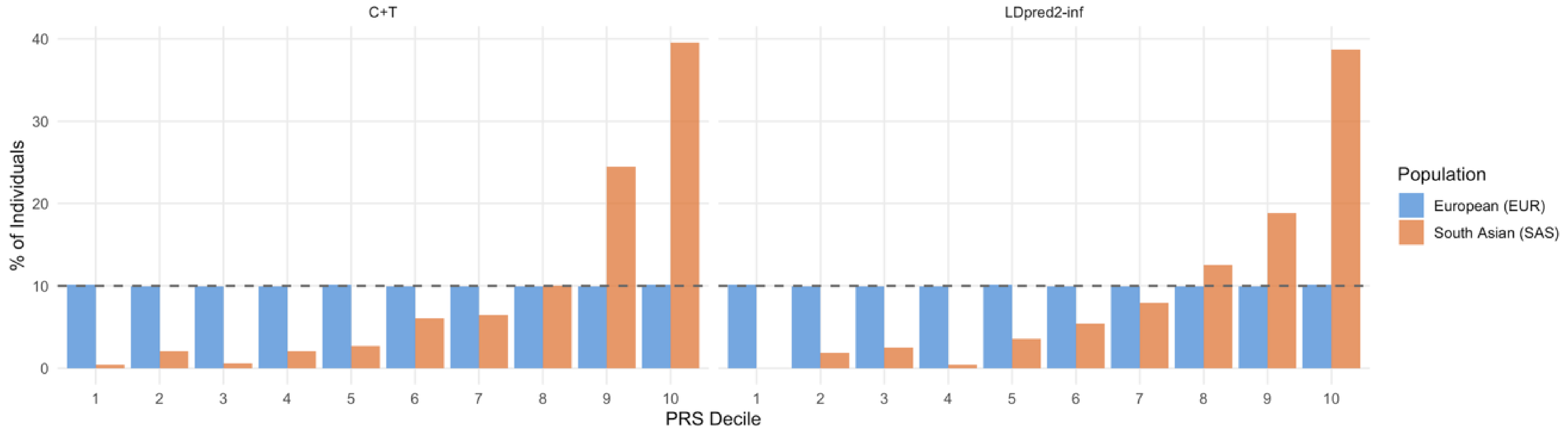


Figure 4. PRS decile distribution: percentage of individuals in each of ten risk deciles

Near-Zero Concordance

between European- and South-Asian-derived individual scores ($r = -0.001$ to 0.047)

~4× Enrichment

of South Asians in the highest-risk decile vs. the 10% expected

High-risk individuals by EUR score ≠ High-risk individuals by CSA score

The Paradox: Shared Biology, Divergent Scores

If the scores disagree so completely — is the underlying disease biology just different?

Individual-Level Scores

$r \approx 0$

near-zero correlation between EUR- and CSA-derived scores

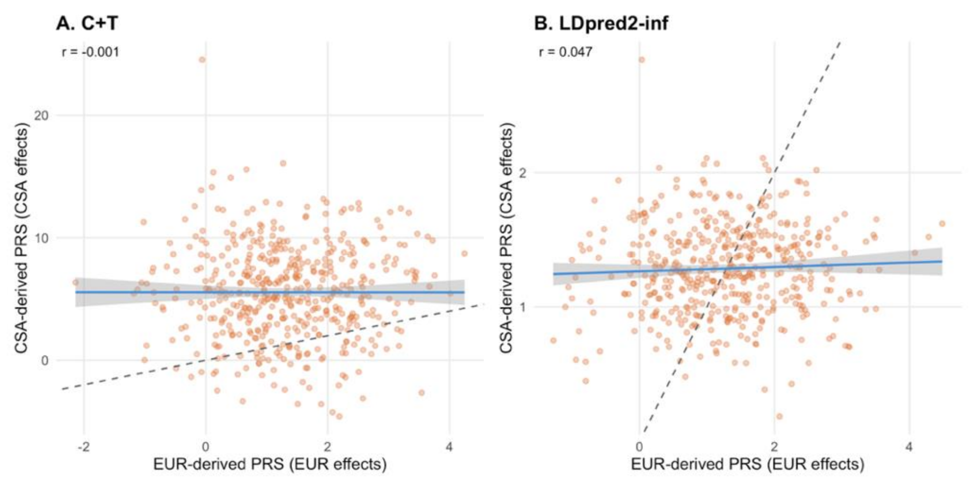


Figure 5. Cross-Ancestry Score Concordance

Looks like a fundamental mismatch...

Genome-Wide Architecture

$rg = 1.27$

$p = 6.7\times10^{-11}$ — indistinguishable from perfect sharing; similar SNP-heritability in both groups (0.036 vs 0.028)

...but the shared biology says otherwise.

Statistical Power Reconciles the Paradox

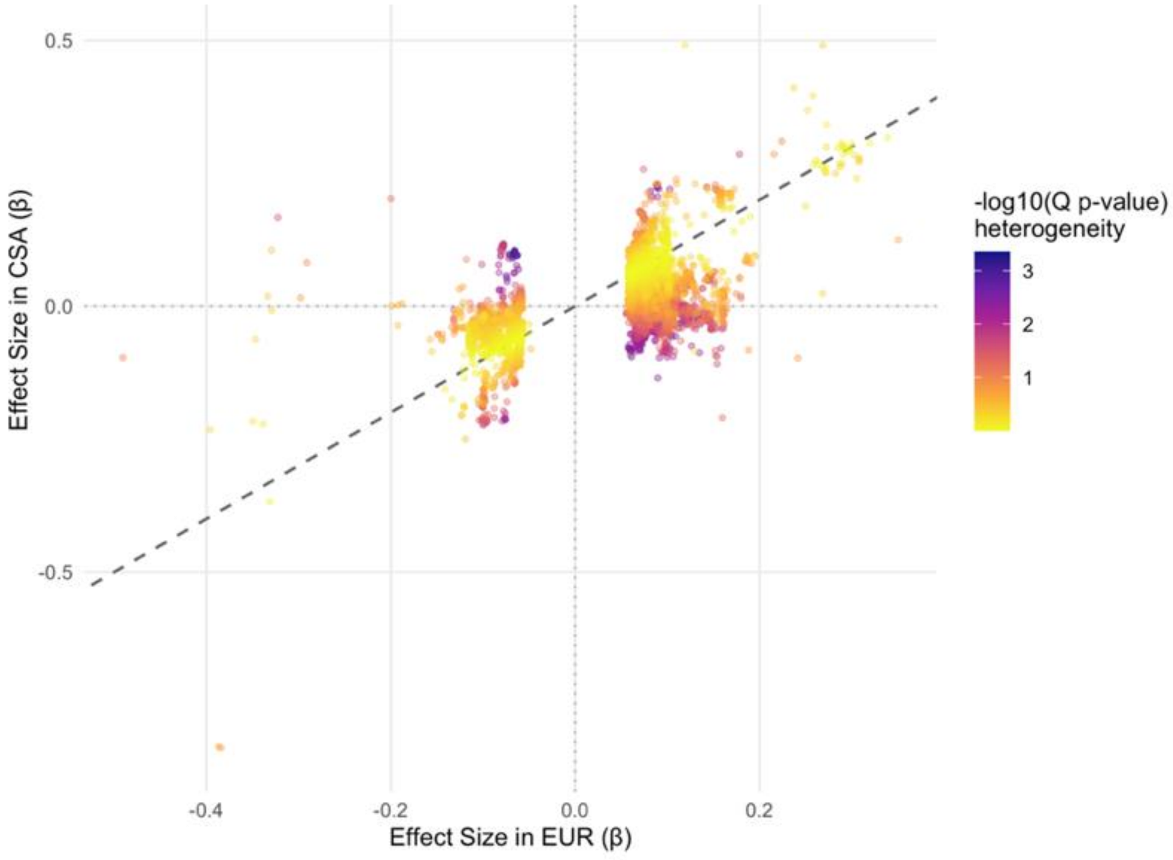


Figure 6. EUR vs. CSA effect sizes at 8,789 EUR genome-wide-significant T2D loci, colored by heterogeneity.

Table 1. Directional Concordance by CSA Signal Strength

CSA |Z|	% Same Direction as EUR
< 1 (weak)	63.7%
1 – 2	71.3%
2 – 3	98.0%
> 3 (strong)	100%

0 of 8,789 genome-wide-significant loci — including TCF7L2, the strongest known T2D gene — showed significant EUR–CSA heterogeneity after correction.

The biology looks shared, not divergent.

Allele Frequency and LD Differences Explain Part of the Gap

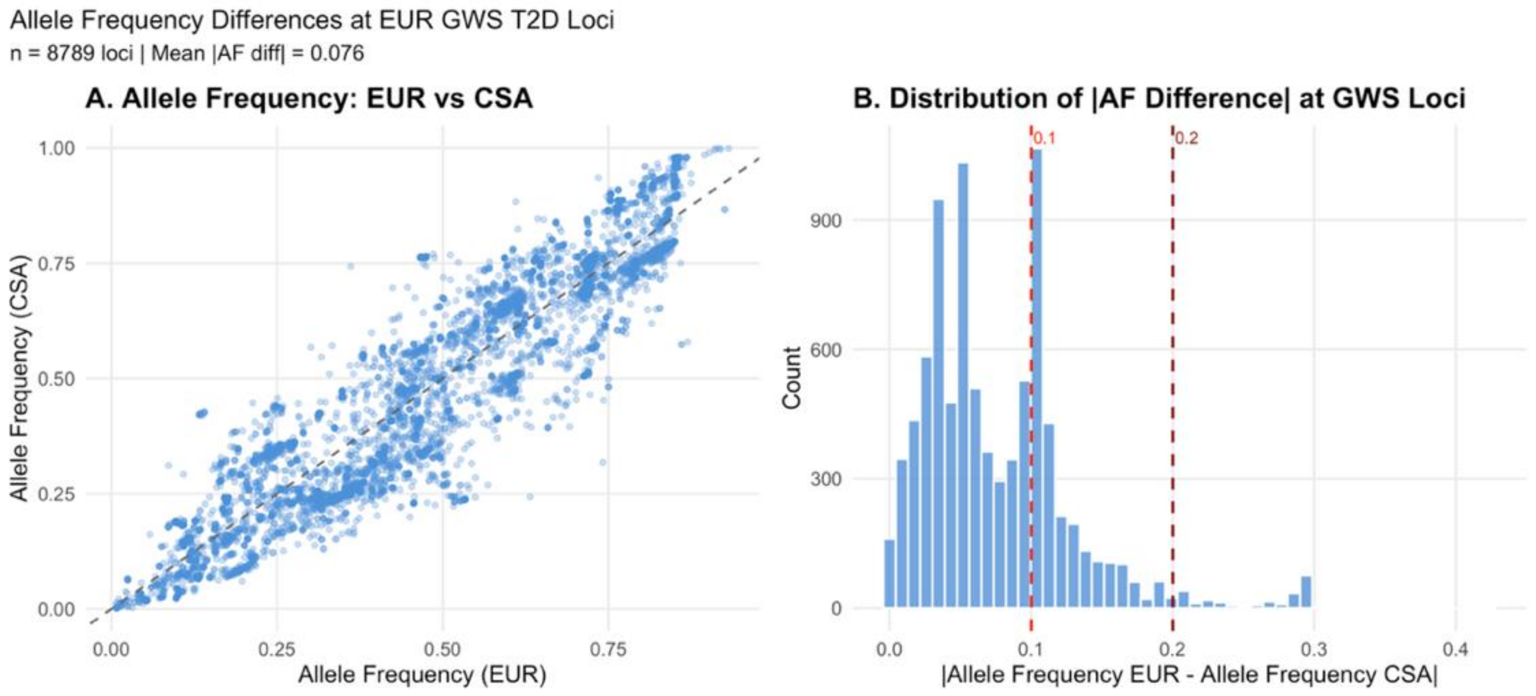


Figure 7. (A) Risk allele frequency, EUR vs. CSA, at 8,789 EUR GWAS T2D loci. (B) Distribution of absolute allele-frequency differences.

Most loci show similar allele frequencies across populations (mean difference = 0.076), but a subset diverge substantially (0.25–0.3). Swapping in a South-Asian-matched LD reference alone reduced the C+T mean shift from 1.32 to 1.19 SD — these mechanisms contribute, but don't fully explain, the miscalibration.

It's Not the Biology — It's the Data

1. **Miscalibration is real and consequential**

   Whole-distribution shifts and 4× over-flagging of high risk in South Asian individuals

2. **But the disease biology is largely shared**

   High genetic correlation (rg=1.27) and zero loci with true EUR–CSA heterogeneity
   (cf. Bulik-Sullivan et al., 2015)

3. **Better algorithms alone didn't fix it**

   LDpred2-inf's more sophisticated LD modeling gave no transferability advantage over C+T
   (cf. Choi et al., 2020; Privé et al., 2021)

4. **The path forward is more data, not more math**

   Expanding South Asian-specific GWAS is the most direct route to equitable T2D risk prediction
   (Martin et al., 2019)

Limitations & What Comes Next

Limitations

- **Genotype-level, not clinical validation:** The 1000 Genomes panel has no phenotype data, so transferability was assessed through genotype-level metrics — score shift, decile enrichment, concordance, heterogeneity, genetic correlation — rather than actual predictive accuracy in diagnosed patients.
- **Small CSA GWAS sample size (1,662 cases):** This is the central limitation of the study — but also arguably its central finding. The underrepresentation itself is the story, not just a caveat on it.
- **"South Asian" is not one population:** The CSA and SAS labels collapse substantial genetic diversity within South Asia (e.g., differences among Pakistani, Bangladeshi, Indian, and Sri Lankan ancestries) that the 1000 Genomes SAS superpopulation doesn't fully capture.

Future Directions

- **Expand CSA-specific GWAS:** Existing efforts — DIAMANTE, SAT2D, the South Asia Biobank — could substantially increase CSA case numbers well beyond the 1,662 used here. *(cf. Mahajan et al., 2022)*
- **Pair genetics with larger phenotyped cohorts:** This would allow direct testing of clinical predictive accuracy, not just genotype-level transferability.
- **Prioritize representation over new methods:** Because miscalibration reflects limited ancestry representation rather than distinct biology, expanding CSA inclusion in genomic studies is the most effective path to equitable T2D PRS — more so than further algorithmic refinement.

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